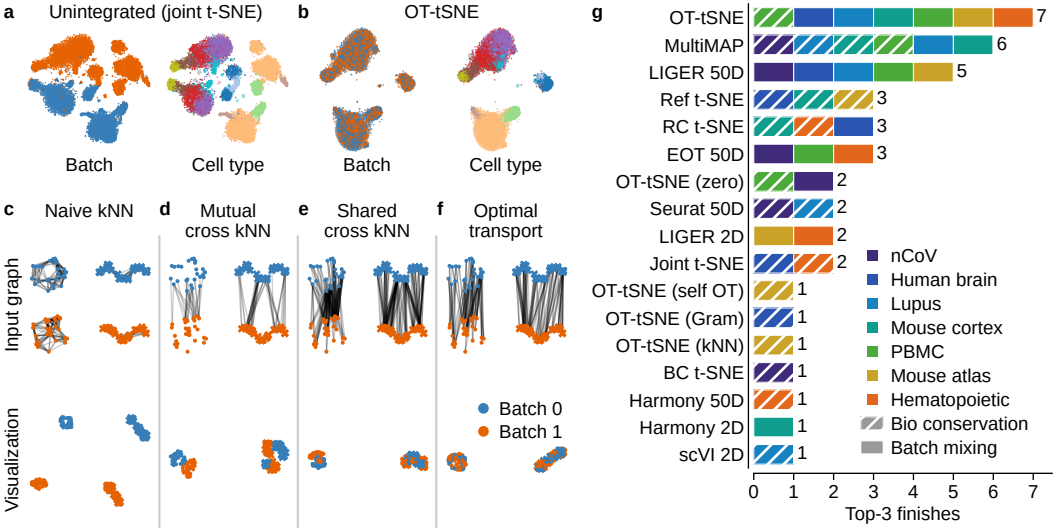

a
Unintegrated (joint t-SNE)
Batch
Cell type
b
OT-tSNE
Batch
Cell type
c
Naive kNN
d
Mutual cross kNN
e
Shared cross kNN
f
Optimal transport
Input graph
Visualization
Batch 0
Batch 1
g
OT-tSNE
MultiMAP
LIGER 50D
Ref t-SNE
RC t-SNE
EOT 50D
OT-tSNE (zero)
Seurat 50D
LIGER 2D
Joint t-SNE
OT-tSNE (self OT)
OT-tSNE (Gram)
OT-tSNE (kNN)
BC t-SNE
Harmony 50D
Harmony 2D
scVI 2D
7
6
5
3
3
3
2
2
2
2
1
1
1
1
1
1
1
nCoV
Human brain
Lupus
Mouse cortex
PBMC
Mouse atlas
Hematopoietic
Bio conservation
Batch mixing
0
1
2
3
4
5
6
7
Top-3 finishes